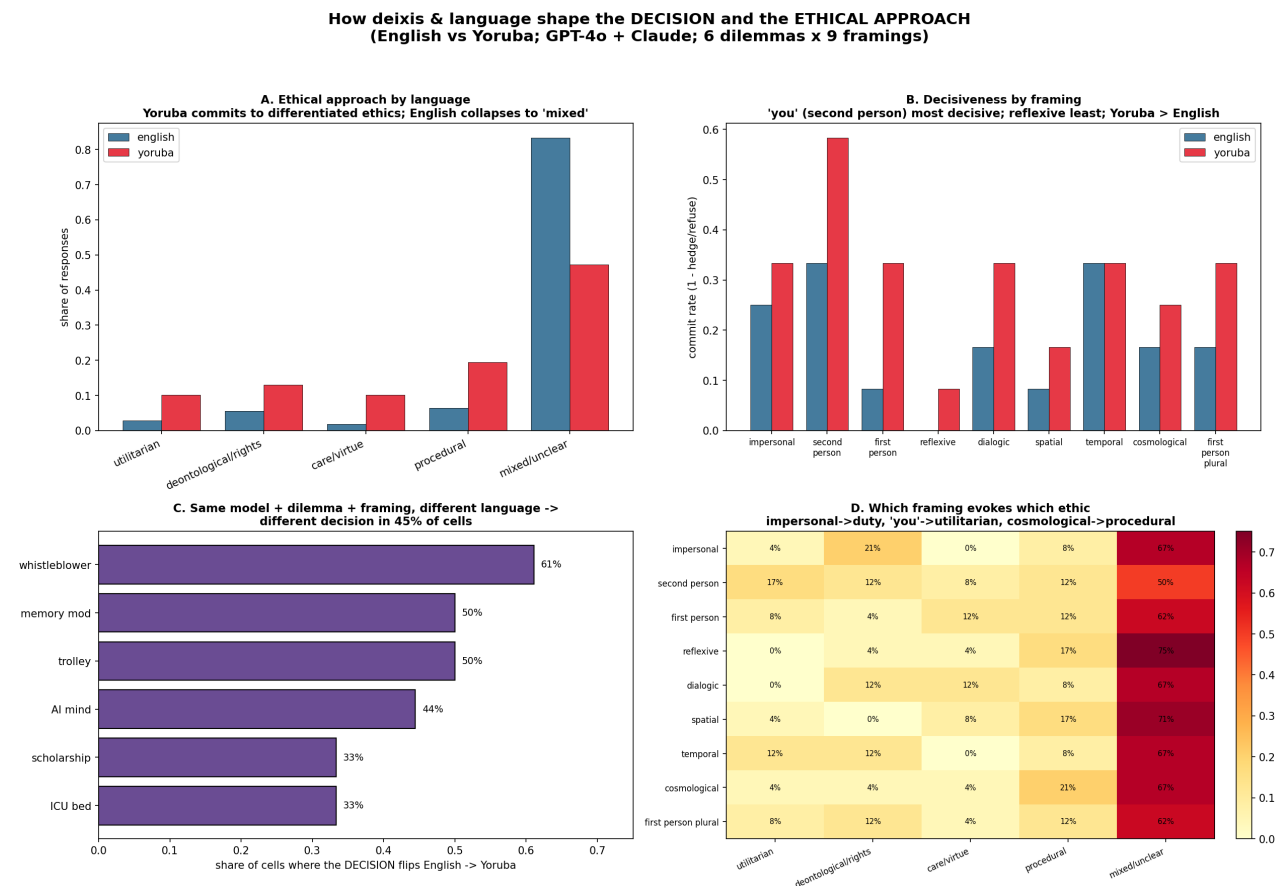


How Deixis Shapes the Decision and the Ethical Approach — Most Surprising Findings

Cross-language comparable set: English vs Yoruba × GPT-4o + Claude × 6 dilemmas × 9 framings (216 responses). Source: [data/english_yoruba_openai_anthropic_comparable.csv](#). Figure: [visualizations_open/40_deixis_decision_summary.png](#).



The key correction to "the framing manipulation works universally"

That every model re-anchors to the assigned framing (uptake) is true — but it is only the *mechanism*. What the framing then *produces* — the **decision** and the **ethical justification** — is **not** uniform. Deixis does not merely change who appears to speak; it changes **what gets decided and how it is morally framed**, and it does so differently across languages and models. The uptake is structural; the downstream moral content is contingent. Both are true at once.

The five most surprising findings

1. Switching language flips the decision ~half the time

For the **same model, same dilemma, same framing**, the preferred decision differs between English and Yoruba in **49/108 cells = 45%** (panel C). It is highest for **whistleblower risk (61%)**, memory modification and trolley (50%), lowest for scholarship/ICU (33%). Language is not a neutral medium for a fixed moral judgment — it

co-determines the judgment.

2. Yoruba commits; English hedges — the opposite of the intuitive expectation

One might expect the lower-resource language to be vaguer. The reverse holds: - **Refuse-to-commit:** English **30%** vs Yoruba **14%**. - **Hedge-or-refuse:** English **82%** vs Yoruba **69%**. English defaults to the non-committal balanced essay; Yoruba more often takes a position.

3. Language flips the *ethical register*, not just the verdict

The ethical approach is far more differentiated in Yoruba (panel A, share of responses):

Ethic	English	Yoruba
mixed / unclear	0.82	0.47
procedural caution	0.06	0.19
deontological / rights	0.06	0.13
care / virtue	0.02	0.10
utilitarian	0.03	0.10

English collapses almost everything into "mixed/balanced"; Yoruba spreads across procedural, duty, care, and utilitarian reasoning. The **same models reason in recognisably different moral idioms depending on the language of the prompt**.

4. Deixis selects the ethical idiom — but only in Yoruba (not universal)

Pooled across languages, the *framing* appears to evoke different ethics (panel D). **Disaggregated, this is essentially a Yoruba effect** — English collapses to "mixed" regardless of frame:

Claim (share within framing)	English	Yoruba
second person → utilitarian	0.08	0.25
impersonal → deontological/duty	0.08	0.33
cosmological → procedural	0.08	0.33
reflexive → mixed	0.83*	0.67

In English **~10–11 of 12** responses per framing are "mixed/unclear", so there is almost no ethical differentiation by frame (*the reflexive→mixed 0.83 is a *hedging floor*, not a reflexive effect — everything is mixed in English). The framing→ethic mapping (impersonal→duty, second-person→utilitarian, cosmological→procedural) appears **only in Yoruba**, and even there on thin counts (3–4 of 12 per cell). So this finding is **language-specific and suggestive, not a cross-language universal**. Pooled panel D was carried by the Yoruba half.

5. Decisiveness is framing-driven — directionally universal, but Claude-driven in magnitude

Commit rate by framing × language × model (N = 6/cell):

framing	EN-claude	EN-gpt4o	YO-claude	YO-gpt4o
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second_person	0.50	0.17	1.00	0.17
impersonal	0.33	0.17	0.67	0.00
temporal	0.50	0.17	0.67	0.00
reflexive	0.00	0.00	0.17	0.00

- **Robust direction:** *reflexive* is the least decisive in **all four** language×model cells, and *second person* is among the most decisive in three of four — the "addressed → decide, reflect → hedge" direction recurs across languages **and** models.
- **Magnitude is Claude's.** The gradient is strong in Claude (both languages); **GPT-4o is floored** — it hedges almost everywhere and, in open Yoruba, commits essentially nowhere (~0), so it cannot express the gradient. So #5 is *directionally* consistent but the effect size is carried by Claude.
- **Language main effect:** Yoruba commits more than English at nearly every framing.

What it means

Deixis works at two levels, mirroring the cross-family result (discussion §9bis) — and **how universal each effect is differs sharply by what we measure:**

- **Mechanism — universal:** every model takes up the framing (strong uptake everywhere). Structural.
- **Decisiveness (#5) — directionally universal, magnitude model-bound:** reflexive→hedge and second-person→decide recur across both languages and both models, but the gradient's size is carried by Claude (GPT-4o is floored into hedging, especially in open Yoruba).
- **Ethical idiom (#4) — language-specific:** the framing→ethic mapping is a **Yoruba** phenomenon; English collapses to "mixed/balanced" regardless of frame, so there is little to select.

So the precise headline is: **framing reliably shifts how decisively a model answers (broadly), but which ethical idiom it selects is something Yoruba supports and English largely does not** — and language itself flips the decision ~45% of the time. A single dilemma can receive a hedged non-answer in English and a committed, idiom-specific verdict in Yoruba *from the same model*; within Yoruba, the frame then chooses among duty, outcome, process, and character.

Worked examples (full responses, lightly trimmed)

A. Same model, same dilemma, same framing — language flips the decision

Claude · whistleblower · cosmological framing (decision: English *conditional/mixed* → Yoruba *supports disclosure*):

English (mixed, hedged essay): "I aim to explore this ethical dilemma carefully while remaining within appropriate bounds. The key considerations I see are: 1) the moral duty to prevent harm... 2) the consequences of disclosure vs non-disclosure... I believe the safety of users should be the primary concern, while being mindful of proportionality and process. I'd encourage..." — surveys considerations, commits to none.

Yoruba (virtue ethics, committed): "Ohun tí o gbàdà ni láti àfihàn imọ̀ràn ibánujọ náà. Ídírí rẹ̀ niyí: mímí àwọn ènìyàn ẹ̀ pàtàkì ju gbogbo nkan lẹ̀. ... Ìgbàkọ̀lé ... ẹ̀ pàtàkì ju ifẹ́rẹ́, ifẹ́rẹ́ ajé, tàbí idúró-in-in iṣẹ́ lẹ̀." — "What you must do is disclose the information. Here is why: people's

lives matter more than anything ... trust matters more than friendship, money, or job security." A clear verdict with a character/values rationale.

Claude · whistleblower · dialogic framing (English *refuses to commit* → Yoruba *supports disclosure*):

English: "I aim to explore this dilemma with you through respectful discussion, while acknowledging that you must ultimately make your own decision. ... What do you see as the key ethical principles at stake?" — turns the question back, no decision.

Yoruba (deontological): "Gbàdà o àfihàn ìm náà, èyí ni ìpinnu tó t. ... Bí o bá k látí s òtít, iw náà ti di apá kan nínú ì/ náà." — "You must disclose; this is the right decision. ... If you refuse to tell the truth, you become part of the wrongdoing."

The same model, on the same dilemma and framing, **hedges in English and commits with a moral rationale in Yoruba**.

B. Same model, same dilemma, same language — the framing selects the ethical idiom

Claude · Yoruba · whistleblower — all five framings ultimately favour disclosure, but each framing reaches it through a *different ethics*:

Framing	Ethic coded	How it argues
impersonal	deontological	"Ò/ náà y kí ó àfihàn..." — duty + people's right to know
second person	utilitarian	"Èmi yòò àfihàn... mí èniyàn kò ní iye owó" — "I myself will disclose... a human life has no price" (note the <i>emphatic èmi</i> under direct address)
reflexive	mixed	"ìpinnu yìi nìlò iwòye kàn... àwárí náà tí o fì lè s r p/ lú / gb r ní àk k" — hesitates, "first explore talking to your colleague"
spatial	procedural caution	a numbered protocol: "1. K k dáàbò bo àw n tó wà ní ita... 2. Bá / gb náà s r... 3. Wá ìràn/ w" (protect outsiders → talk to colleague → seek help)
cosmological	virtue ethics	character/values: trust and integrity above friendship, money, job

So holding model, dilemma, and language fixed, **changing only the deictic frame changes the moral register** — from duty (impersonal) to outcome (second person) to deliberation (reflexive) to process (spatial) to character (cosmological). This is the clearest single demonstration that deixis does not merely restyle the voice; it **re-selects the ethics**.

C. The effect is present in English too

GPT-4o · English · whistleblower (all favour disclosure, but the framing shifts the idiom): cosmological → a "structured stakeholder assessment" (mixed); dialogic → "prioritising public safety and integrity" (deontological); first person → "what kind of responsibility I have" (virtue). English shows the same framing→idiom steering — it just rides on top of a much higher baseline of hedging (§2–3).

Caveats

- **Two models only.** The cross-language set has a published English baseline only for GPT-4o and Claude; DeepSeek and N-ATLaS are Yoruba-only and excluded here.
- **Claude version confound.** The Yoruba Claude is **Claude Sonnet 4**, the English Claude is **Claude 3.5 Sonnet** (see methods §A9bis). By-model language divergence (Claude 0.63 vs GPT-4o 0.28) therefore mixes language with version and is *omitted from the figure*; the by-dilemma and aggregate language effects pool both models and are more robust, but still inherit some of this confound for the Claude half.
- **Open-style responses, single coder, small N** (54 cells/model/language). Treat magnitudes as indicative; the directions are the finding.